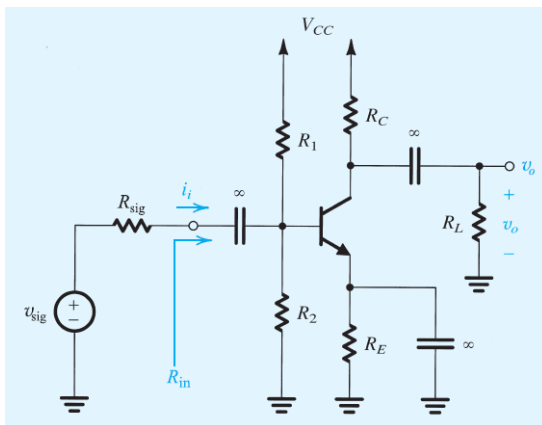


Homework 11

Topic: BJTs Small Signal Analysis

- For the common-emitter amplifier shown below, let $V_{CC}=15\text{ V}$, $R_1=27\text{ k}\Omega$, $R_2=15\text{ k}\Omega$, $R_E=2.4\text{ k}\Omega$, and $R_C=3.9\text{ k}\Omega$. The transistor has $\beta=100$.
 - Calculate the dc bias current I_C .
 - If the amplifier operates between a source for which $R_{sig}=2\text{ k}\Omega$ and a load of $2\text{ k}\Omega$, replace the transistor with its hybrid- π model, and find the values of R_{in} , and the overall voltage gain v_o/v_{sig} .
 - Find R_{out} .
 - Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants
 - Make a rough sketch of the Bode plot (magnitude only).



- The amplifier below consists of two identical common-emitter amplifiers connected in cascade. Observe that the input resistance of the second stage, R_{in2} , constitutes the load resistance of the first stage.
 - For $V_{CC}=15\text{ V}$, $R_1=100\text{ k}\Omega$, $R_2=47\text{ k}\Omega$, $R_E=3.9\text{ k}\Omega$, $R_C=6.8\text{ k}\Omega$, and $\beta=100$, determine the dc collector current and dc collector voltage of each transistor.
 - Draw the small-signal equivalent circuit of the entire amplifier and give the values of all its components.
 - Find R_{in1} and v_{b1}/v_{sig} for $R_{sig}=5\text{ k}\Omega$.
 - Find R_{in2} and v_{b2}/v_{b1} .
 - For $R_L=2\text{ k}\Omega$, find v_o/v_{b2} .
 - Find the overall voltage gain v_o/v_{sig} .
 - Find R_{out} .
 - Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants
 - Make a rough sketch of the Bode plot (magnitude only).

